The Journal of Open Source Software

cogsworth: A Gala of COSMIC proportions combining binary stellar evolution and galactic dynamics

Tom Wagg [1,2], **Katelyn Breivik** [3], **Mathieu Renzo** [2], **and Adrian M. Price-Whelan** [4]

1 Department of Astronomy, University of Washington, Seattle, WA, 98195 **2** Center for Computational Astrophysics, Flatiron Institute, 162 Fifth Ave, New York, NY, 10010, USA **3** McWilliams Center for Cosmology and Astrophysics, Department of Physics, Carnegie Mellon University, Pittsburgh, PA 15213, USA **4** University of Arizona, Department of Astronomy & Steward Observatory, 933 N. Cherry Ave., Tucson, AZ 85721, USA

DOI: 10.xxxxxx/draft

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Editor:

Submitted: 06 September 2024
Published: unpublished

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Summary

We present `cogsworth`, an open-source Python tool for producing self-consistent population synthesis and galactic dynamics simulations. With `cogsworth` one can (1) sample a population of binaries and star formation history, (2) perform rapid (binary) stellar evolution, (3) integrate orbits through the galaxy and (4) inspect the full evolutionary history of each star or compact object, as well as their positions and kinematics. We include the functionality for post-processing hydrodynamical zoom-in simulations as a basis for galactic potentials and star formation histories to better account for initial spatial stellar clustering and more complex potentials. Alternatively, several analytic models are available for both the potential and star formation history. `cogsworth` can transform the intrinsic simulated population to an observed population through the joint application of dust maps, bolometric correction functions and survey selection functions.

Statement of need

The majority of stars are born in binaries and multiple star systems (e.g., Duchêne & Kraus (2013); Moe & Di Stefano (2017); Offner et al. (2023)), a large subset of which will exchange mass at some point in their lifetime (e.g., Podsiadlowski et al. (1992); Sana et al. (2012); de Mink et al. (2014)). These massive stars play a critical role in the formation and evolution of galaxies as a result of their feedback (e.g., Dekel & Silk (1986); Hopkins et al. (2012); Nomoto et al. (2013); Somerville & Davé (2015); Naab & Ostriker (2017)). However, binary evolution remains uncertain, with many parameters such as common-envelope efficiency, mass transfer efficiency, angular momentum loss due to mass transfer and the mean magnitude of supernova natal kicks unconstrained across several orders of magnitude (e.g., Janka (2012); N. Ivanova et al. (2013); Katsuda et al. (2018); Natalia Ivanova et al. (2020); Röpke & De Marco (2023); Marchant & Bodensteiner (2023)).

Single massive stars are not expected to migrate far from their birth location before reaching core-collapse due to their short lifetimes ($\lesssim 50$,Myr, e.g., Zapartas et al. (2017)). However, binary stars may disrupt after an initial supernova event, ejecting the secondary star from the system with its orbital velocity (e.g., Blaauw (1961); Eldridge et al. (2011); Renzo et al. (2019)). Thus, close massive binaries that disrupt can lead to the displacement of secondary stars significantly farther from star-forming regions. The present-day positions and kinematics of massive stars and binary products are therefore strongly impacted by changes in binary physics that alter the separation prior to supernova. This means that comparing simulations of

positions and kinematics of stars and compact objects to observations will enable constraints on binary stellar evolution parameters.

The use of positions and kinematics as tracers of binary evolution has been considered in the past. Recent work has shown the importance of accounting for the galactic potential, which can change the velocity of kicked objects (e.g. Disberg et al. (2024)). It is also important to consider the inclination or timing of a supernova kick relative to the galactic orbit, since, for example, a kick out of the galactic plane at an object's highest galactic vertical position will have a strong effect on its final position. Failing to consider impacts from both a galactic potential and kicks (i.e. velocity impulses) will lead to misleading conclusions regarding the final spatial distributions of the population. Some studies have considered using the Galactic potential at the present-day position of objects to place a lower limit on the peculiar velocity at birth and constrain supernova kicks (Atri et al., 2019; Repetto et al., 2012, 2017; Repetto & Nelemans, 2015), but the accuracy of this method is debated (Mandel, 2016). Other work has considered the impact of the Galactic potential for individual special cases, rather than at a population level. For example, Evans et al. (2020) considered the orbits of hyper-runaway candidates evolving through the Milky Way potential, whilst Neuhäuser et al. (2020) developed software for tracing the motion of stars to investigate the recent nearby supernovae that ejected ζ Ophiuchi.

There is a clear need for an open-source tool that provides the theoretical infrastructure for making predictions for the positions and kinematics of massive stars and compact objects, placing these systems in the context of their host galaxy and its gravitational potential. `cogsworth` fulfils this need, providing a framework for self-consistent population synthesis and galactic dynamics simulations. The code is applicable to a wide range of binary products, both common and rare, from walkaway and runaway stars to X-ray binaries, as well as gravitational-wave and gamma-ray burst progenitors.

Acknowledgements

We gratefully acknowledge many fruitful discussions with Julianne Dalcanton and Eric Bellm that resulted in several helpful suggestions. TW acknowledges valuable conversations with Matt Orr and Chris Hayward regarding the FIRE simulations, and with Alyson Brooks and Akaxia Cruz regarding the ChaNGa simulations. TW thanks the Simons Foundation, Flatiron Institute and Center for Computational Astrophysics for running the pre-doctoral program during which much of this work was completed. The Flatiron Institute is supported by the Simons Foundation. TW and KB acknowledge support from NASA ATP grant 80NSSC24K0768.

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